

1.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
    }
    else if (pid == 0)
    {
        // Child process
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        // Only runs if exec fails
        printf("Command execution failed!\n");
    }
    else
    {
        // Parent process
        printf("Parent PID: %d\n", getpid());

        wait(NULL);

        printf("Child process finished.\n");
    }

    return 0;
}
```

```
ananya_roy@Ananya:~$ nano practical1.c
ananya_roy@Ananya:~$ gcc practical1.c
ananya_roy@Ananya:~$ ./a.out
Enter a Linux command: ls
Parent PID: 1625
Child PID: 1627
Parent PID: 1625
a.out  a9    exec2.c    linux_task1  permission_lab  pipeex2.c    skill3.c
a10    abc   execex1.c  linux_task2  pipe1.c         pipeex3.c    student_portal
a11    b1    execex2.c  linux_task3  pipe2.c         practical1.c  xyz
a2.c   b2    execvex3.c linux_task4  pipe3.c         readwr.c     z7
a7     dirz  f1         open1.c      pipeex1         skill1.c
a8     ex1   f1.c       ossp         pipeex1.c       skill2.c
Child process finished.
```

Linux terminal commands

```

ananya_roy@Ananya:~$ uname
Linux
ananya_roy@Ananya:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:        0-11
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) 5 120U
CPU family:                  6
Model:                      186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    3
BogoMIPS:                    4992.02
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
                             pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb r
                             dtscp lm constant_tsc rep_good nopl xtopology tsc_reliable non
                             stop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16
                             pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes
                             xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ss
                             bd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fs
                             gsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx
                             smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx
                             _vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri mov
                             dir64b fsrm md_clear serialize ibt flush_lld arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       288 KiB (6 instances)
  L1i:                       192 KiB (6 instances)
  L2:                         7.5 MiB (6 instances)
  L3:                        12 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-11
Vulnerabilities:
  Gather data sampling:      Not affected
  Ghostwrite:                Not affected
  Indirect target selection: Not affected
  Itlb multihit:             Not affected
  L1tf:                      Not affected
  Mds:                       Not affected
  Meltdown:                  Not affected
  Mmio stale data:           Not affected

```

```
ananya_roy@Ananya:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro

ananya_roy@Ananya:~$ ps
  PID TTY          TIME CMD
  316 pts/0    00:00:00 bash
 1695 pts/0    00:00:00 ps

ananya_roy@Ananya:~$ top
top - 15:00:12 up 50 min,  1 user,  load average: 0.00, 0.03, 0.00
top - 15:02:00 up 51 min,  1 user,  load average: 0.02, 0.03, 0.00
Tasks: 25 total,  1 running, 24 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7750.6 total, 7075.5 free,   559.8 used,   271.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,    0.0 used. 7190.8 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24540 15540 11532 S   0.0   0.2   0:01.55 systemd
    2 root        20   0   3180   2204 2072 S   0.0   0.0   0:00.16 init-syste+
top - 15:02:08 up 52 min,  1 user,  load average: 0.02, 0.03, 0.00
Tasks: 27 total,  1 running, 26 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7750.6 total, 7073.5 free,   561.8 used,   271.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,    0.0 used. 7188.8 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24540 15540 11532 S   0.0   0.2   0:01.55 systemd
    2 root        20   0   3180   2204 2072 S   0.0   0.0   0:00.16 init-systemd(Ub
    6 root        20   0   3180   2044 1940 S   0.0   0.0   0:00.00 init
   49 root        19  -1  50400 16856 15552 S   0.0   0.2   0:00.44 systemd-journal
   81 systemd+    20   0  22416 14480 11992 S   0.0   0.2   0:00.20 systemd-resolve
   89 root        20   0  35364 12232  9156 S   0.0   0.2   0:02.02 systemd-udev
  110 root        20   0   2888   1948 1820 S   0.0   0.0   0:00.13 chronyd-starter
  111 root        20   0   4472   3176 2928 S   0.0   0.0   0:00.02 cron
  112 message+   20   0   8824   5440 4716 S   0.0   0.1   0:00.14 dbus-daemon
  128 root        20   0  44096 29652 14512 S   0.0   0.4   0:00.59 networkd-dispat
  163 root        20   0  18436  9432  8184 S   0.0   0.1   0:00.24 systemd-logind
  173 syslog      20   0  220548 5904 4624 S   0.0   0.1   0:00.17 rsyslogd
  227 _chrony     20   0  20424  6388 5608 S   0.0   0.1   0:00.15 chronyd
  234 _chrony     20   0  12096  2464 1820 S   0.0   0.0   0:00.01 chronyd
  282 root        20   0   5492   2752 2564 S   0.0   0.0   0:00.01 agetty
  284 root        20   0 123460 32776 18168 S   0.0   0.4   0:00.39 unattended-upgr
  313 root        20   0   3184   1108  980 S   0.0   0.0   0:00.00 SessionLeader
  314 root        20   0   3200   1248 1108 S   0.0   0.0   0:00.49 Relay(316)
  316 ananya_+    20   0   6268  5552 3864 S   0.0   0.1   0:00.31 bash
  317 root        20   0   8644   5516 4676 S   0.0   0.1   0:00.01 login
  381 ananya_+    20   0  22344 13584 10952 S   0.0   0.2   0:00.19 systemd
  385 ananya_+    20   0  22916  4128 2124 S   0.0   0.1   0:00.00 (sd-pam)
  411 ananya_+    20   0   6136  5456 3864 S   0.0   0.1   0:00.04 bash
  536 polkitd     20   0 306444  8444 7632 S   0.0   0.1   0:00.22 polkitd
 1705 ananya_+    20   0   8340   6148 3900 R   0.0   0.1   0:00.09 top
 1720 root        20   0  35368  6444 3332 S   0.0   0.1   0:00.00 (udev-worker)
 1721 root        20   0  35368  6444 3332 S   0.0   0.1   0:00.00 (udev-worker)
```

Command	Role/Purpose
uname	Displays the os being used.
lscpu	Displays CPU architecture, cores, threads, and processor
lsblk	Displays storage devices, partitions, and their sizes.
ps	Displays currently running processes and their PIDs.
top	Provides real-time information about processes and CPU/memory

2.

```
#include<stdio.h>
#include<unistd.h>
#include<fcntl.h>

int main()
{
    int fd1;
    int fd2;
    char buf[100];
    size_t n;

    fd1=open("a7", O_RDWR);
    fd2=open("z7", O_RDWR);

    printf("Files successfully opened and created.\n");
    printf("%d\n",fd1);
    printf("%d\n",fd2);

    n=read(fd1,buf,30);
    //printf("%s\n",buf);
    write(fd2,buf,n);

    return 0;
}
```

```
ananya_roy@Ananya:~$ nano open3.c
ananya_roy@Ananya:~$ gcc open3.c
ananya_roy@Ananya:~$ ./a.out
Files successfully opened and created.
3
4
```

Strace:

```

ananya_roy@Ananya:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7fff7b2ac178 /* 25 vars */) = 0
brk(NULL)                               = 0x56caeab0b000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x743a6935e000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27889, ...}) = 0
mmap(NULL, 27889, PROT_READ, MAP_PRIVATE, 3, 0) = 0x743a69357000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x743a69324000
mmap(0x743a6932b000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x743a6932b000
mmap(0x743a6934c000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x743a6934c000
mmap(0x743a69353000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x743a69353000
mmap(0x743a69355000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x743a69355000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x743a692f6000
mmap(0x743a692fa000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x743a692fa000
mmap(0x743a6931e000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x743a6931e000
mmap(0x743a69322000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x743a69322000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x743a691d0000
mmap(0x743a691e2000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x743a691e2000
mmap(0x743a69282000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x743a69282000
mmap(0x743a692f4000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x743a692f4000
close(3)                                = 0

```

3.

```

#include <unistd.h>
#include <stdio.h>

int main()
{
    int pid;

    pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("Child Process id is = %d\n", getpid());
    }
    else if(pid > 0)
    {
        printf("Parent Process\n");
        printf("Parent process id is = %d\n", getppid());
    }
    else if(pid < 0)
    {
        printf("Failed process\n");
    }

    return 0;
}

```

```

ananya_roy@Ananya:~$ nano practical2.c
ananya_roy@Ananya:~$ gcc practical2.c
ananya_roy@Ananya:~$ ./a.out
Parent Process
Parent process id is = 316
Child Process
Child Process id is = 1978

```

Experiment:

```
ananya_roy@Ananya:~$ sleep 100 &
[5] 679
ananya_roy@Ananya:~$ ps -p 679 -o pid,stat,cmd
  PID STAT CMD
   679 S    sleep 100
ananya_roy@Ananya:~$ cat /proc/679/status
Name:      sleep
Umask:    0022
State:    S (sleeping)
Tgid:     679
Ngid:     0
Pid:      679
PPid:     392
TracerPid: 0
Uid:      1000    1000    1000    1000
Gid:      1000    1000    1000    1000
FDSize:   256
Groups:   4 24 27 30 46 100 1000
NSTgid:   679
NSpid:    679
NSpgid:   679
NSsid:    392
Kthread:  0
VmPeak:   16112 kB
VmSize:   16112 kB
VmLck:    0 kB
VmPin:    0 kB
VmHWM:    7712 kB
VmRSS:    7712 kB
RssAnon:  1224 kB
RssFile:  6488 kB
RssShmem: 0 kB
VmData:   288 kB
VmStk:    132 kB
VmExe:    4832 kB
VmLib:    3240 kB
VmPTE:    68 kB
VmSwap:   0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:  1
SigQ:     1/30959
SigPnd:   0000000000000000
ShdPnd:   0000000000000000
SigBlk:   0000000000000000
SigIgn:   0000000000000000
SigCgt:   0000000000000000
CapInh:   0000000000000000
```